



Question: What makes good fire wood?

- Density
- Water content

Denser and drier firewood will burn better and produce more heat.

Value builds progressively

1. Data = fuel

- raw data without inherent meaning

2. Knowledge = fire

- Information/ knowledge drawn from the raw data. The quality and condition of the raw data controls the quality and condition of our analytics.
- processing and understanding the data
- Finding patterns and relationships
- Creating models and operational definitions

We transform the raw data to fuel our knowledge. "Knowledge is power" and we use it to benefit ourselves and others.

3. Value = heat for warmth and cooking

- applying insights to create value
- real-world applications and decisions

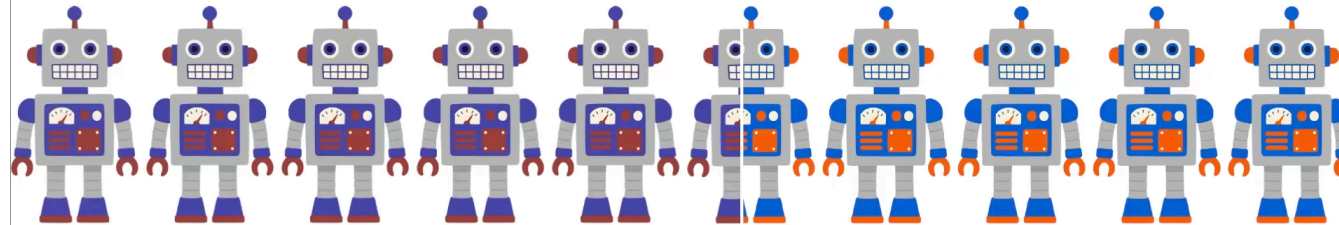
We need the right conditions, tools, and knowledge to transform data into value.

"Chance favours the prepared mind."

"Garbage in, garbage out"

Data is a resource → Make data available
Knowledge is power → Make data meaningful
Wisdom is value → Make data valuable

DATA ON DATA



0%

Jonker, A., & Aquino, J. (2026, February 10). *What is dirty data?* IBM. <https://www.ibm.com/think/topics/dirty-data>

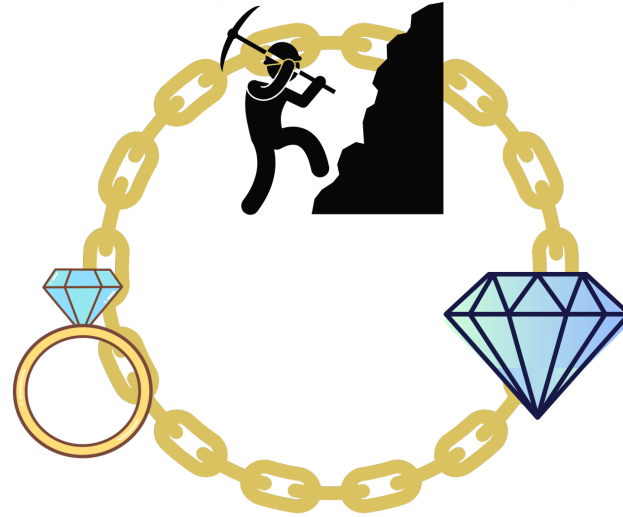
Why Data Governance Matters

- Defines who owns data, how it's handled, and what rules it follows
- **Analogy:** Governance = "air traffic control" for data
- Orchestrates access, quality standards, and compliance

> 54% of executives say effective data governance/management is a top priority (IBV survey)

GOVERNANCE IN THE DATA LIFE CYCLE

EXTRACT, TRANSFORM, LOAD



Characteristics of Good Data

- Falsifiable
- Reproducible

Data Governance Framework

Defined Roles & Responsibilities

- Governance council/steering committee sets strategy
- Data owners: accountable for quality in their domain
- Data stewards: handle day-to-day quality management

Clear Policies & Standards

- Documented rules for formatting, naming, access, protection

Auditing & Monitoring Procedures

- Ongoing audits to assess quality, compliance, and standards adherence

BAD DATA

STALE DATA

DARK DATA

DIRTY DATA

Jonker, A., & Krantz, T. nd. *What is data quality?* IBM. <https://www.ibm.com/think/topics/data-quality>

Bad data refers to information that compromises decision-making because it is inaccurate, incomplete, inconsistent, outdated, duplicate, invalid or biased.

Stale data is information that's outdated, misaligned with current conditions or no longer fit for its intended purpose. Also referred to as stale information or old data, it represents one of the most pervasive and under-addressed challenges in modern data management.

Dark data is the information that organizations accumulate but often never use for analytics or decision-making.

Dirty data is information that is inaccurate, invalid, incomplete or inconsistent, making it unreliable for business use.

Bad Data

- often at the source of collection
 - guesses or dubious data
 - a portion of the problem that goes unreported
 - cleverly-worded
- Federal gov't survey on AI
- no negative options

Stale data

- degrading quality over time

Dark data

File Drawer Effect

No relationship or effect ≠ Bad data

Replication Crisis

"Many Labs Replication Project" (Klein et al., 2013)

- mixed results
- unethical



BAD DATA



STALE DATA

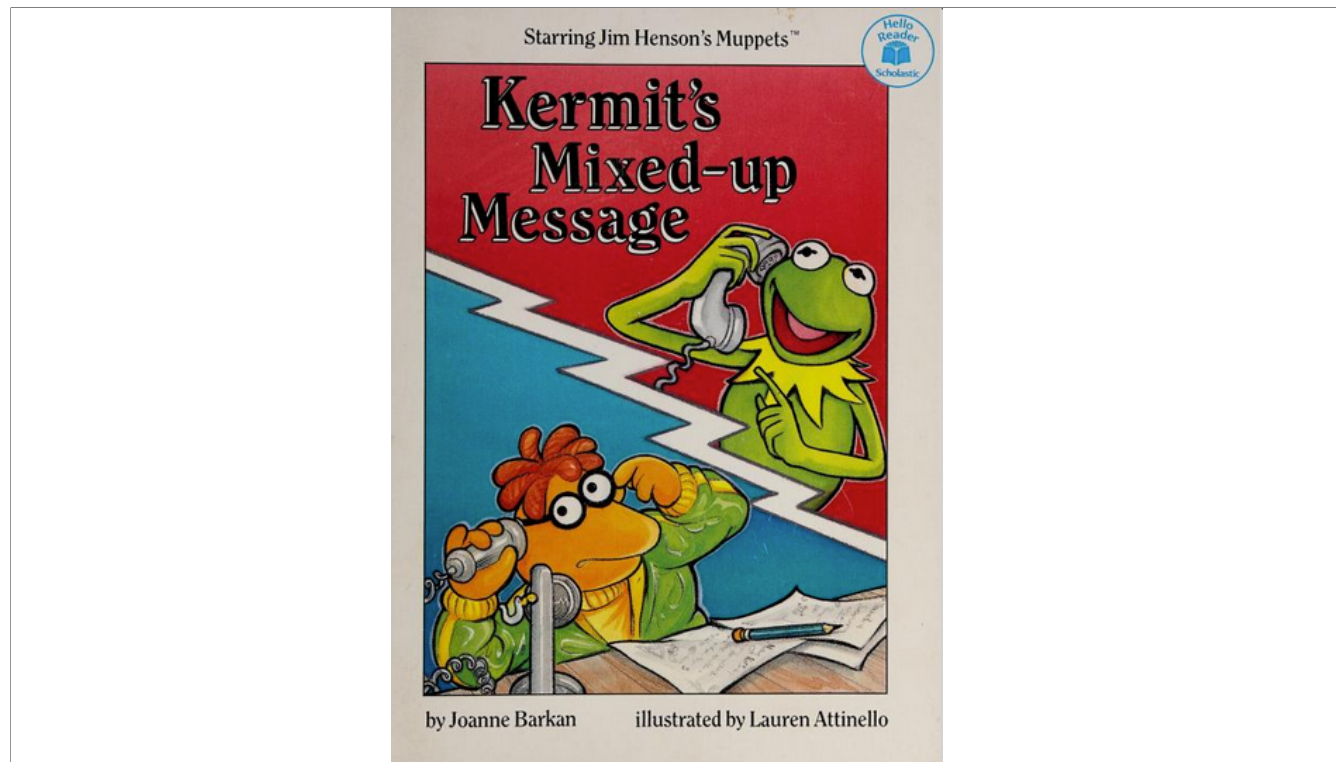


DARK DATA

DIRTY DATA



Jonker, A., & Krantz, T. nd. *What is data quality?* IBM. <https://www.ibm.com/think/topics/data-quality>



- The Muppets are rehearsing for the big show, so you can't hear what Kermit is saying over the phone:
 - Lights out in the great hall, but the other Muppets think:
 - Termites in Miss Piggy's ball gown
 - Gonzo will fall from the tightrope
 - Kermit doesn't like Fozzi's new jokes

Dirty Data is a form of miscommunication because it is inaccurate, invalid, incomplete, and inconsistent.
Dirty data can distort analytics, increase scaling errors, degrade AI model performance, lead to poor decision-making.

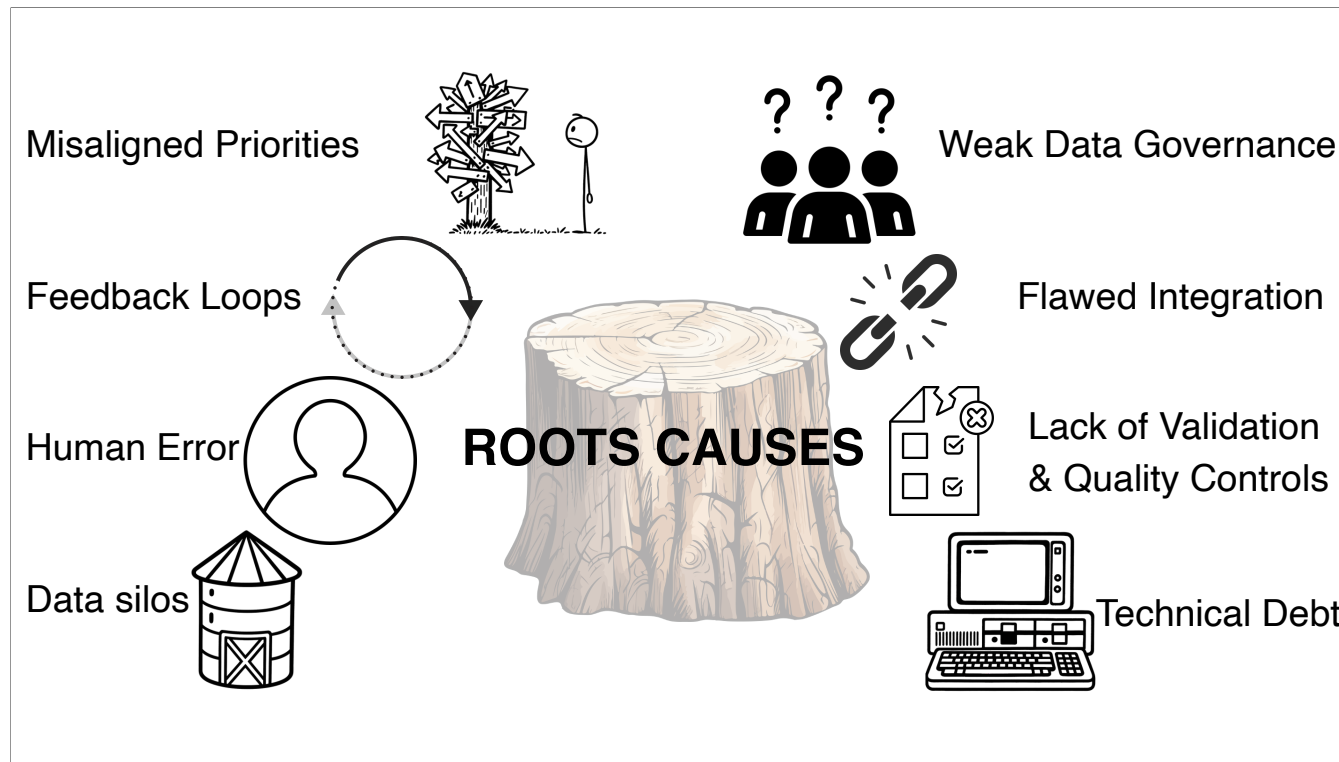


Dirty data can occur any where in the data life cycle and it compounds problems

- outdated data, duplicate records -> poor decision making & planning
- Incomplete data -> ineffective marketing/sales/ customer experience
- Lower confidence in analysis -> delayed decisions
- Time consuming data cleaning & reconciliation
- Increased dependency on highly skilled individuals
- Slower innovation
- Reduced return on investment/lost competitive advantage



IBM survey, 2025
25%+ of organizations estimate annual losses **exceeding \$5 million USD** due to poor data quality (Forrester)



Root Cause: Human Error

- Manual entry is error-prone (repetition, time pressure, cognitive load)
- Results in typos, transposed characters, misread sources, copy-paste mistakes
- Systematic errors multiply quickly → costly cleanup

Root Cause: Data Silos & Weak Governance

Data Silos

- Fragmented info across departments
- Isolated datasets → duplicate & misaligned records
 - Difficult to resolve

Weak Data Governance

- No centralized oversight or defined ownership
- Leads to conflicting formats, naming conventions, definitions

Root Cause: Integration & Technical Debt

Flawed Data Integration

- Schema mismatches, faulty transformations, incomplete transfers
- Increased by cloud/hybrid architectures

Technical Debt

- Outdated data models, limited validation, brittle legacy interfaces
- Forces manual workarounds; introduces unflagged outliers

Root Cause: Validation, Priorities & ML Loops

Lack of Validation & Quality Controls

- No range checks, format enforcement, required fields, uniqueness constraints
- Errors enter silently, propagate downstream

Misaligned Priorities

- Speed/volume rewarded over accuracy → rising error rates

Machine Learning Feedback Loops

- Models trained on flawed/biased data

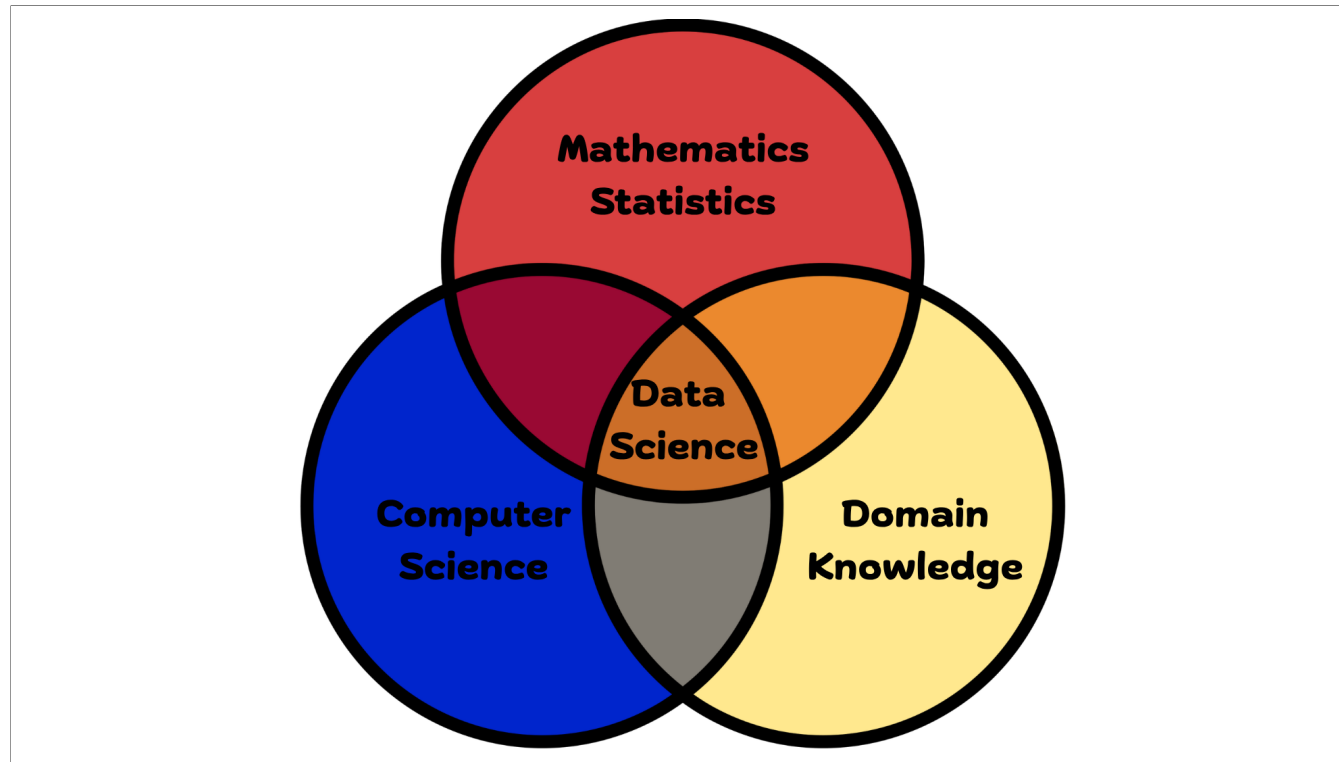
- Outputs reintegrated as inputs without validation → amplification



Not one method to remediation, every situation is unique

Knowledge synthesis:

- mathematical and statistical knowledge
- Computer science
- Domain knowledge



Knowledge synthesis

- each area shares a portion with the others and data science is a subsection of all three
- You need all 3 domains to find meaning and to create value from raw data

Data science integrates three main domains:

1. Computer Science

- Algorithms and programming
- Data structures
- System design

2. Mathematics/Statistics

- Statistical analysis
- Mathematical modeling
- Probability theory

3. Domain Knowledge

- Field-specific expertise
- Understanding context
- Problem definition

The field requires understanding:

- Where data comes from
- What questions to ask
- How to apply insights

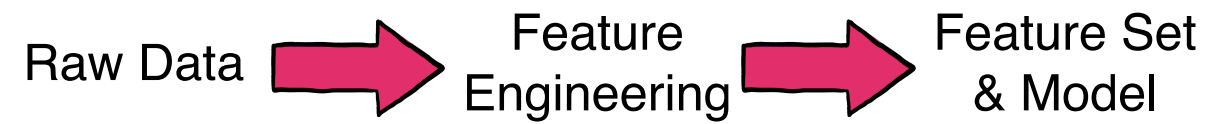
HOW TO CLEAN DIRTY DATA

- Feature engineering “the creative bridge between raw data and models”
 - Feature engineering and data integration are crucial skills that bridge raw data and meaningful insights, enabling more effective decision-making across various domains.
 - It creates a reliable and unified data set. It transforms raw, messy data into actionable insights.
- Examples: format timestamps for readability
- improves model compatibility

Dirty Data & AI

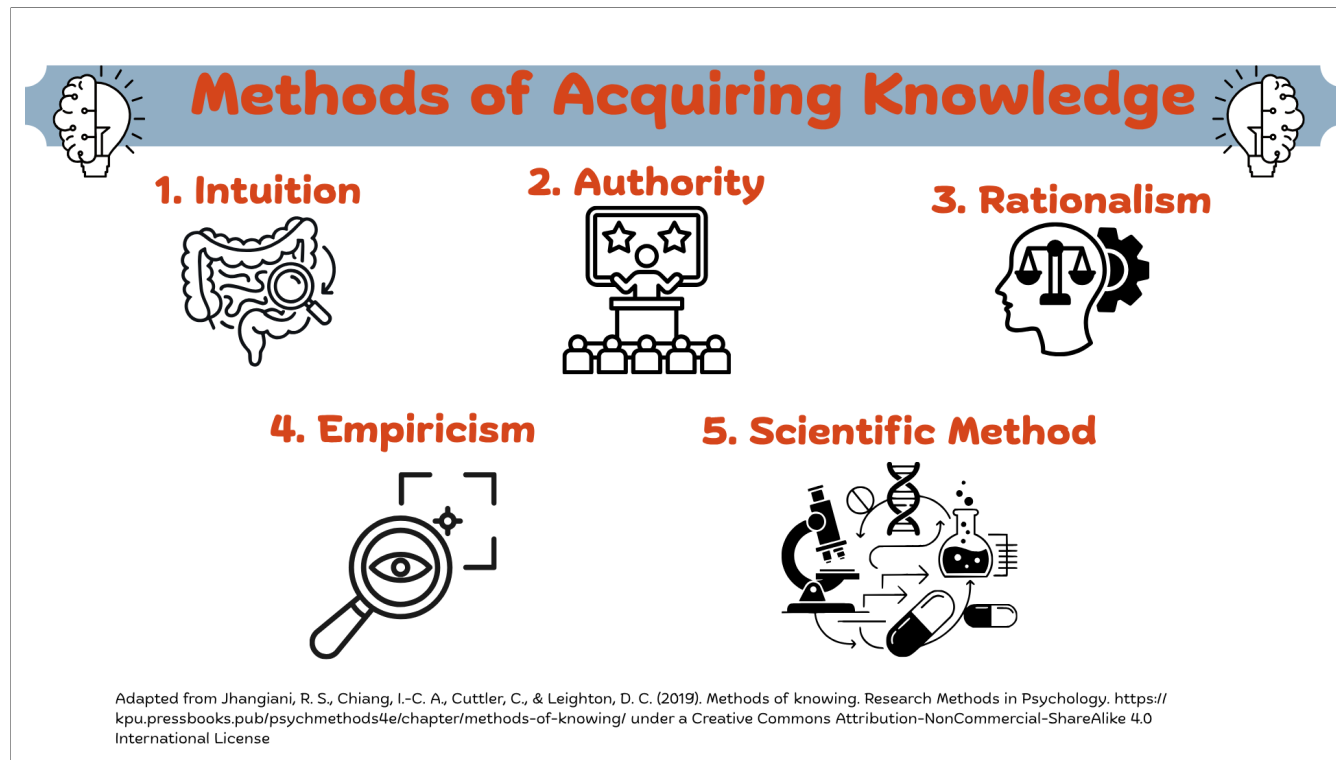
- AI/LLMs learn statistical patterns at scale
- Errors & biases in training data → flawed, misleading outputs
- **Gartner prediction:** Through 2026, organizations will abandon **60%** of AI projects unsupported by AI-ready data
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- **Payoff of clean data:** Enterprises with widely trusted data achieve **nearly double** the ROI on AI investments (IBV research)

HOW TO CLEAN DIRTY DATA



- *Fix errors*
- *Standardize*
- *Deduplicate*
- *Impute*
- *Readability*

DOMAIN KNOWLEDGE + CREATIVITY



Methods of Acquiring Knowledge

1. Intuition
2. Authority
3. Rationalism
4. Empiricism
5. Scientific Method

Intuition

Feels true

>"When we use our intuition, we are relying on our guts, our emotions, and/or our instincts to guide us. Rather than examining facts or using rational thought, intuition involves believing what feels true."

Authority

Because I say so

> One of the most common methods of acquiring knowledge involves accepting new ideas because an authority figure says its true. Much of the information we acquire is through authority because we don't have time to question and independently research every piece of knowledge. But we can learn to evaluate the credentials of authority figures, to evaluate the methods they used to arrive at their conclusions, and evaluate whether they have any reasons to mislead us.

Rationalism

Logic and reasoning

Faulty logic

> Rationalism involves using logic and reasoning to acquire new knowledge. Using this method premises are stated and logical rules are followed to arrive at sound conclusions. The problem with this method is that if the premises are wrong or there is an error in logic then the conclusion will not be valid.

Empiricism

Observation and Experience

Anecdotal vs Systematic Observation

> Empiricism involves acquiring knowledge through observation and experience. We are limited in what we can experience and observe and our senses can deceive us. Moreover, our prior experiences can alter the way we perceive events. Nevertheless, empiricism is at the heart of the scientific method. Science relies on observations. But not just any observations, science relies on structured observations which is known as systematic empiricism.

Scientific Method

Testing ideas in controlled conditions

> The scientific method is a process of systematically collecting and evaluating evidence to test ideas and answer questions. While scientists may use intuition, authority, rationalism, and empiricism to generate new ideas they don't stop there. Scientists go a step further by using systematic empiricism to make careful observations under various controlled conditions in order to test their ideas, using rationalism to arrive at valid conclusions.

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